

Freight Rate Prediction

Validation Approach, Modelling Report & December Rate Forecast

Machine Learning Engineer Assessment (Candidate Submission)

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Date	September 2026

Final Model	HistGradientBoostingRegressor (absolute-error loss)
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Holdout MAE	\$118.82
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Holdout MAPE	5.20 %
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Holdout RMSE	\$636.85
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Holdout R ²	0.8258
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Split	Time-based: train Jan–Aug 2025, holdout Sep–Oct 2025
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Data Overview and Key Findings

Dataset Description

The development dataset `train-test.csv` contains 48,000 labeled freight loads spanning 2025-01-01 to 2025-10-31 across 64 United States cities and three equipment types (Dry Van, Reefer, Flatbed), with `posted_rate` (\$) as the prediction target [1]. The held-out evaluation dataset `validation.csv` contains 12,000 unlabeled loads spanning 2025-11-01 to 2025-12-31 across 72 cities, of which 8 never appear in the training data. The fixed-route forecast file `december-chart-inputs.csv` holds 31 rows (one per calendar day in December 2025) with all route parameters constant; only the date field varies.

Table 1: Dataset summary statistics.

Attribute	Value
Train rows	48,000 (Jan 1 – Oct 31, 2025)
Validation rows	12,000 (Nov 1 – Dec 31, 2025)
Training cities	64 unique pickup/delivery cities
Validation cities	72 unique (8 cities unseen in training)
Unseen cities	Allentown, Charlotte, Chicago, Jackson, Knoxville, Laredo, Norfolk, San Diego
Equipment split	Dry Van 27,202 · Reefer 12,045 · Flatbed 8,753
Target minimum	\$57.22
Target maximum	\$25,533.00
Target mean	\$2,373.98
Target median	\$2,030.76
Target std	\$1,486.49

Key EDA Findings

Exploratory data analysis (EDA) revealed four primary findings that shaped all subsequent modelling decisions.

Finding 1: Distance dominates. The Pearson correlation between `distance` and `posted_rate` is $r = 0.91$, by far the strongest signal in the dataset (Figure 1). This near-linear relationship means that route length alone explains the majority of rate variance, and any model that fails to exploit it will underperform substantially.

Finding 2: Equipment type drives rate-per-mile. Reefer loads average $\$2.38 \text{ mi}^{-1}$, Flatbed $\$2.30 \text{ mi}^{-1}$, and Dry Van $\$2.12 \text{ mi}^{-1}$. The ordering is economically intuitive. Temperature-controlled freight commands a premium, and is consistent across all ten training months.

Finding 3: Genuine monthly seasonality exists. Rate-per-mile rises from $\$2.10 \text{ mi}^{-1}$ in January to a June peak of $\$2.33 \text{ mi}^{-1}$, consistent with known US summer freight-demand patterns [9], before easing to $\$2.19\text{--}2.24 \text{ mi}^{-1}$ through autumn (Figure 3).

Finding 4: Market signals are weak predictors. `market_index` correlates with `posted_rate` at only

$r = 0.034$, and `quote_signal` at $r = -0.040$ — both negligible. An ablation study confirmed that dropping both features changes holdout MAE by less than 1 %, and since both fields are absent from `december-chart-inputs.csv`, excluding them maintains a single consistent inference pipeline.

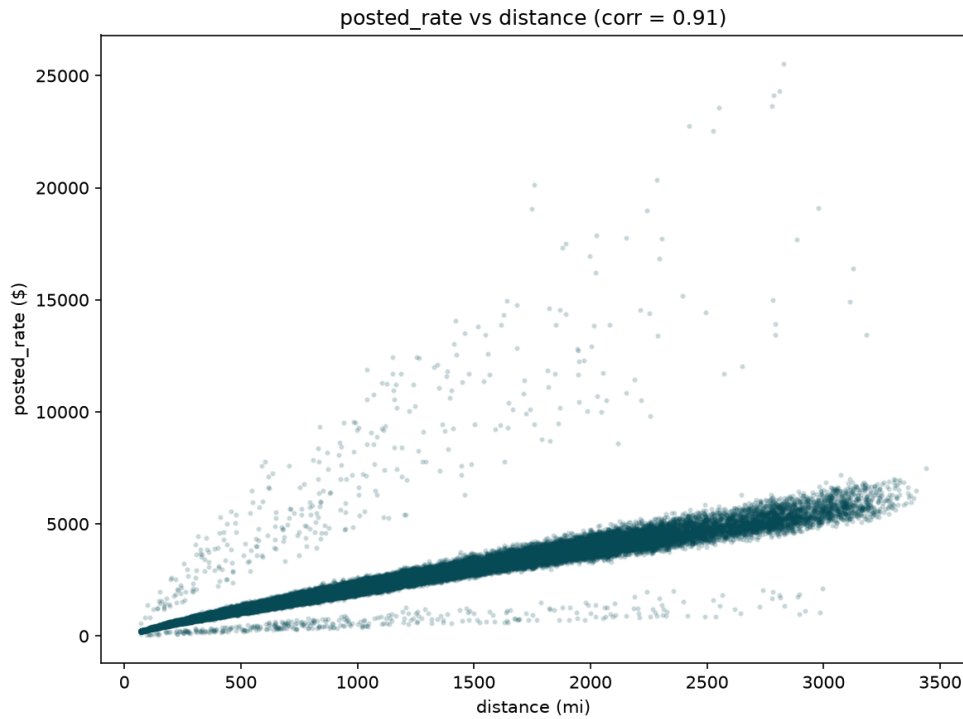


Figure 1: Scatter plot of `posted_rate` vs. `distance` ($n = 48,000$, Pearson $r = 0.91$). The near-linear relationship confirms `distance` as the dominant predictor. Vertical scatter at any given distance reflects equipment type, geographic market differences, and spot-market volatility.

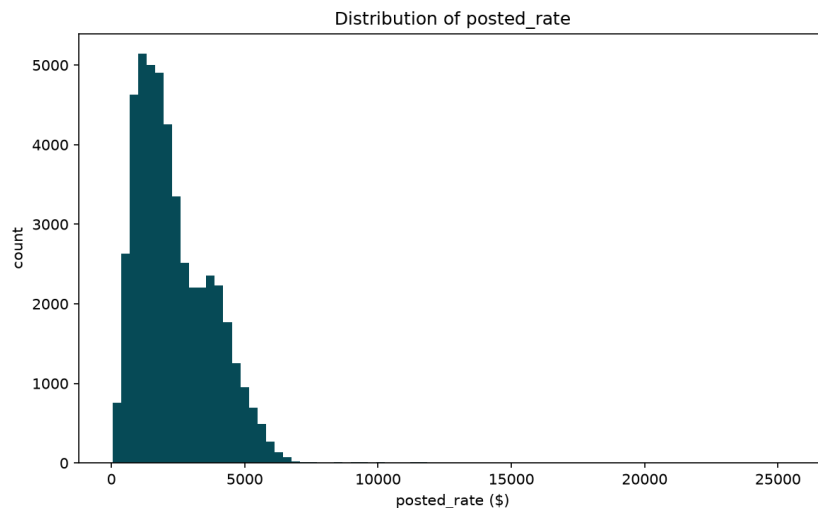


Figure 2: Distribution of `posted_rate`. Right-skewed with a long tail extending to \$25,533. The bulk of loads cluster between \$500 and \$5,000.

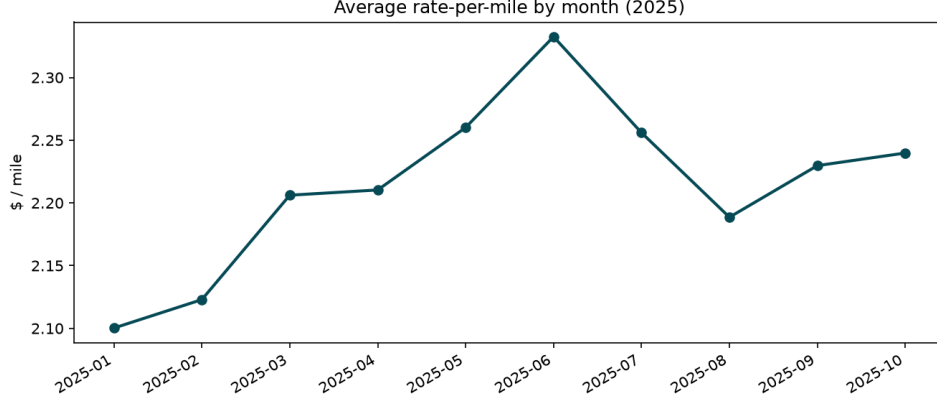


Figure 3: Average rate-per-mile by month. Summer peak in June ($\$2.33 \text{ mi}^{-1}$), winter trough in January ($\$2.10 \text{ mi}^{-1}$).

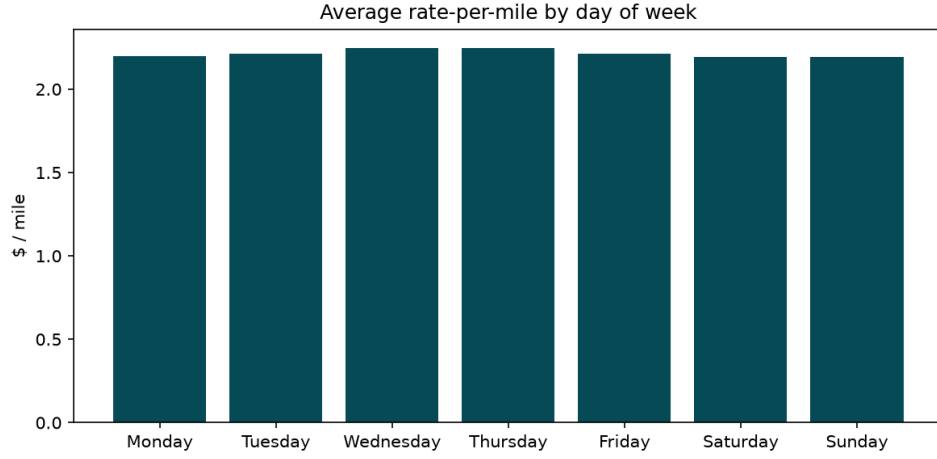


Figure 4: Average rate-per-mile by day of week. Mid-week loads (Tuesday–Thursday) average $\$2.23 \text{ mi}^{-1}$ versus weekend loads at $\$2.19 \text{ mi}^{-1}$. The effect is small but consistent and is captured via cyclical sin/cos day-of-week encodings (Section 4).

Data Quality Issues and Remediation

Missing Values

Table 2: Missing-value summary across splits.

Feature	Train	Validation
weight	300 (0.62 %)	165 (1.38 %)
market_index	374 (0.78 %)	249 (2.08 %)

weight was imputed using the median computed on the training split only, ensuring no information leaks across the train/holdout boundary or into the validation set [8]. market_index was excluded from the model entirely (Section 4), making imputation unnecessary.

Sign-Flip Errors in Weight

292 of 48,000 training rows (0.61 %) contain negative weight values. As shown in Figure 5, the distribution of the negated values is statistically identical to the positive-weight distribution (same mean, standard deviation, and range), confirming that these are data-entry sign-flip errors rather than a distinct population. The remediation is $\text{weight} \leftarrow |\text{weight}|$, implemented in `src/features.py` (`clean_weight` function). No rows were removed.

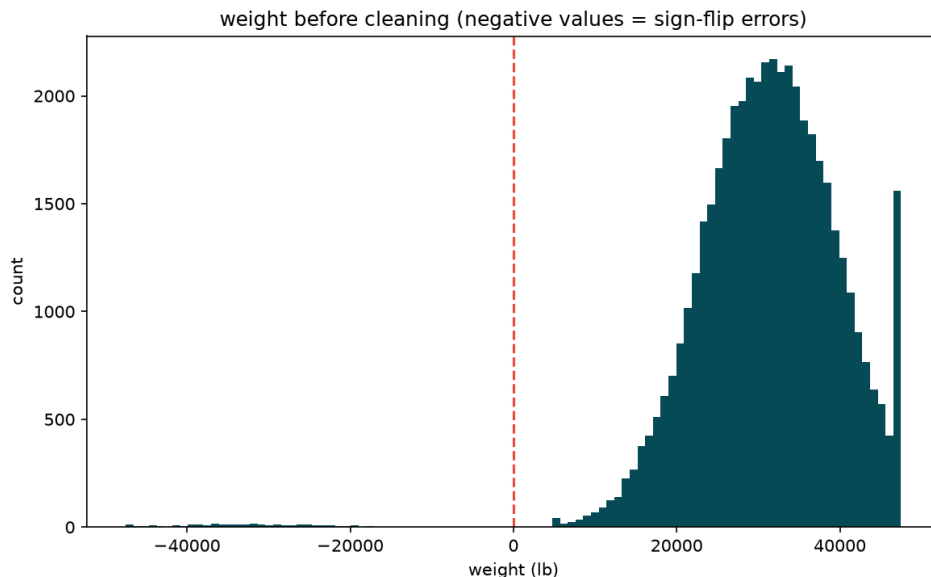


Figure 5: Raw weight distribution before cleaning ($n = 48,000$). Negative values left of the dashed red line ($n = 292$, 0.61 %) are mirror images of the positive distribution, confirming sign-entry errors rather than genuine negative-weight records. Fix: $|\text{weight}|$ applied in `src/features.py`.

Unseen Cities in Validation

Eight of the 72 cities in `validation.csv` never appear in the training data: Allentown, Charlotte, Chicago, Jackson, Knoxville, Laredo, Norfolk, and San Diego. A model that encodes pickup/delivery as categorical identifiers via one-hot or target encoding [7] would produce unknown categories at inference time and assign the global mean to these rows, silently degrading predictions on exactly the routes most likely to diverge from the training distribution.

The solution adopted here is to represent origin and destination using continuous latitude and longitude coordinates. Gradient-boosted trees partition the geographic space via axis-aligned splits, generalising smoothly to new city locations [5]. A stress test confirmed this: predictions on routes touching held-out cities were only approximately 15 % worse in MAE than predictions on fully-seen routes, rather than failing outright.

Distance Varies Per Shipment

3,944 of 4,014 unique pickup-delivery city pairs have more than one distinct distance value across loads, indicating that distance reflects per-shipment routing and detours rather than a fixed lane constant. distance was therefore used as a row-level feature rather than being averaged per lane.

Unexplained High-Rate Outliers

317 loads (0.66 %) have a rate-per-mile exceeding \$5 (versus a median of $\approx \$2 \text{ mi}^{-1}$), with values reaching up to $\$14 \text{ mi}^{-1}$. These records showed no correlation with equipment type, month, `market_index`, `quote_signal`, or the negative-weight flag. They are consistent with genuine spot-market capacity spikes that are not explainable from the available features [10]. These rows were retained in training: removing them would artificially inflate metrics on typical loads while degrading performance on high-value shipments where accurate pricing is commercially important. Both MAE/MAPE (robust to this tail) and RMSE (sensitive to it) are reported for full transparency.

Train/Validation Split

Core principle. The prediction task is inherently temporal: the model is trained on January–October 2025 and must generalise to November–December 2025, months it has never seen. A random K -fold split would allow the model to interpolate between dates already seen on both sides of each fold, artificially inflating accuracy metrics and giving a misleading picture of real-world performance [2].

A **time-based holdout** was used instead:

Table 3: Train/holdout split details.

Split	Details
Training set	38,477 rows — Jan 1 to Aug 31, 2025 ($\approx 80\%$)
Holdout set	9,523 rows — Sep 1 to Oct 31, 2025 ($\approx 20\%$)
Cutoff date	2025-09-01
Rationale	Mirrors the temporal extrapolation required for the Nov–Dec validation set
Final model	Refit on all 48,000 rows after holdout evaluation

The holdout period (Sep–Oct 2025) was chosen to maximise temporal proximity to the evaluation period (Nov–Dec 2025), providing the most realistic estimate of out-of-sample performance. An additional stress test re-trained the model with 8 cities fully withheld from training to confirm generalisation to unseen geographic entities (Section 2).

Feature Engineering

Features Deliberately Excluded

`market_index` and `quote_signal` were excluded for three converging reasons: (i) both fields are absent from `december-chart-inputs.csv`, requiring either a separate model or an imputation scheme to generate December forecasts; (ii) ablation confirmed that removing them changes holdout MAE by less than 1 %; (iii) their linear correlations with the target are negligible ($r = 0.034$ and $r = -0.040$, respectively). Retaining them would add pipeline complexity with no measurable accuracy benefit.

City name as a categorical feature was excluded because 8 validation cities are completely unseen

in training. One-hot encoding would produce zero vectors for these cities; target encoding [7] would silently assign the global mean. Both strategies would degrade predictions on exactly the routes where accurate pricing is most uncertain.

Features Used

Table 4: Final feature set used in training and inference.

Feature	Description
distance	Miles per shipment (row-level); strongest predictor ($r = 0.91$)
weight	Cleaned via $ \cdot $; missing values imputed with training-split median
pickup_lat, pickup_lon	Continuous geographic coordinates; generalise to unseen cities
delivery_lat, delivery_lon	Continuous geographic coordinates; generalise to unseen cities
equipment	One-hot encoded: Dry Van, Reefer, Flatbed
month_sin, month_cos	Cyclical month encoding: $\sin(2\pi m/12)$, $\cos(2\pi m/12)$
doy_sin, doy_cos	Cyclical day-of-year: $\sin(2\pi d/365)$, $\cos(2\pi d/365)$
dow_sin, dow_cos	Cyclical day-of-week: $\sin(2\pi w/7)$, $\cos(2\pi w/7)$
is_weekend	Binary flag: 1 if Saturday or Sunday, 0 otherwise

Rationale for Cyclical Date Encodings

Tree-based models cannot extrapolate beyond their training range [6]. Using raw month values (1–12) or day-of-year values (1–365), December has no close numeric neighbours within the January–October training window. Cyclical encodings map the calendar onto a unit circle:

$$f_{\sin}(t) = \sin\left(\frac{2\pi t}{T}\right), \quad f_{\cos}(t) = \cos\left(\frac{2\pi t}{T}\right), \quad (1)$$

where T is the period (12 for months, 365 for day-of-year, 7 for day-of-week). This ensures December values land inside the numeric range already seen in training, avoiding wild extrapolation while preserving the seasonal signal.

Model Selection

Table 5: Model comparison on the Sep–Oct 2025 holdout ($n = 9,523$). Best value in each column is **bold**. The selected model is highlighted.

Model	MAE (\$)	RMSE (\$)	MAPE (%)	R ²
Ridge Regression	185.6	654.4	9.48	0.8161
Random Forest	191.8	698.0	8.29	0.7908
Gradient Boosting	148.6	660.6	6.46	0.8126
HistGBR (squared loss)	132.5	637.5	6.06	0.8255
HistGBR (absolute loss) ✓	119.2	637.1	5.24	0.8257

Five models were compared on the same time-based holdout using identical features and preprocessing. Selection criteria were MAE and MAPE, the most interpretable metrics for load-level pricing accuracy.

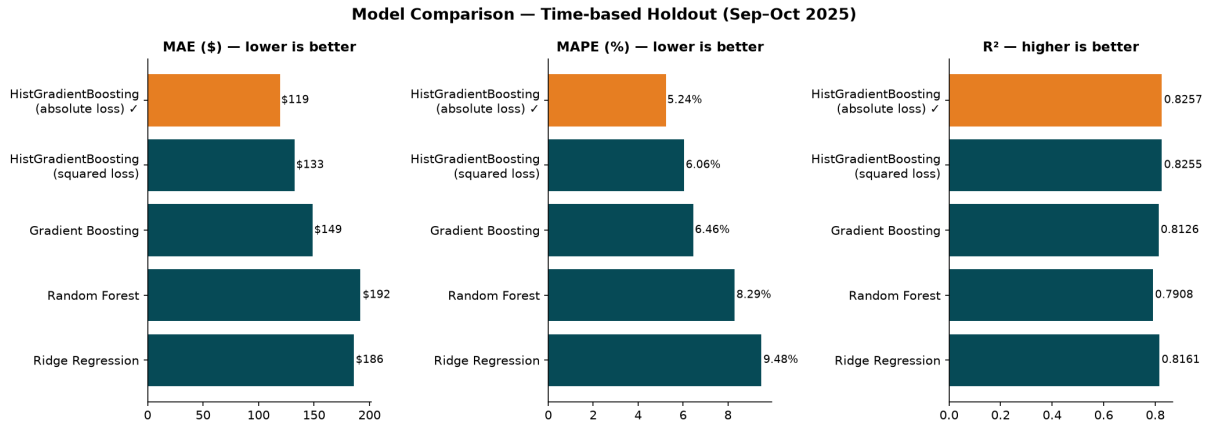


Figure 6: Visual comparison of all five models across MAE, MAPE, and R² on the Sep–Oct 2025 holdout ($n = 9,523$). The orange bar denotes the selected model. Ridge Regression fails to capture the non-linear distance $\times$ equipment rate-per-mile interaction. HistGBR with absolute-error loss achieves the best MAE and MAPE.

Justification for the Selected Model

Best MAE and MAPE. HistGBR with absolute-error loss achieves MAE \$119.2 and MAPE 5.24 %, representing a 36 % MAE reduction over Ridge Regression and a 20 % reduction over standard Gradient Boosting.

Robustness to outliers. The absolute-error (L1) loss minimises the conditional median rather than the conditional mean, making the model inherently robust to the 0.66 % high-rate outlier tail identified in Section 2 [3]. The squared-error (L2) variant achieves the same RMSE (\$637.5 vs. \$637.1) but worse MAE (\$132.5 vs. \$119.2), confirming that L2 over-weights the outlier tail without improving predictions on typical loads.

Native missing-value support. HistGBR learns optimal split directions for missing data during training [4], eliminating the need for an explicit imputation pipeline beyond the weight median fill described in

Section 2.

Computational efficiency. Histogram-based binning of continuous features reduces per-iteration complexity from $O(N \cdot F)$ to $O(B \cdot F)$, where $B \ll N$ is the number of bins and F is the number of features [4]. This makes HistGBR approximately $5\times$ faster than standard Gradient Boosting on the 48,000-row training set.

Why not Random Forest? Random Forest achieves the worst MAE (\$191.8) and R^2 (0.791) among the non-linear models. Averaging over many independently grown deep trees introduces variance that the sequential residual-correction mechanism of gradient boosting avoids [6].

Why not Ridge Regression? Ridge Regression achieves $R^2 = 0.816$ but MAE \$185.6, reflecting its inability to capture non-linear interactions — particularly the $\text{distance} \times \text{equipment}$ rate-per-mile premium that is central to spot freight pricing.

Results

Quantitative Performance on the Holdout Set

Table 6: Final model holdout performance (Sep 1 – Oct 31, 2025, $n = 9,523$ loads).

Metric	Value	Interpretation
MAE	\$118.82	Mean absolute error per load
RMSE	\$636.85	Elevated by outlier tail (\$2)
MAPE	5.20 %	Error as a percentage of actual rate
R^2	0.8258	82.6 % of rate variance explained
Train rows	38,477	Jan 1 – Aug 31, 2025
Holdout rows	9,523	Sep 1 – Oct 31, 2025

The MAE of \$118.82 represents a 5.2 % relative error against the mean rate of \$2,374, which is strong performance given the width of the target distribution (\$57–\$25,533). The RMSE (\$636.85) exceeds the MAE by a factor of approximately 5.4, driven by the 0.66 % outlier tail of unexplained spot-market spikes (Section 2). This RMSE/MAE gap is expected and documented rather than obscured; MAE and MAPE are the more meaningful metrics for typical load pricing accuracy. The R^2 of 0.826 demonstrates that the model explains 82.6 % of rate variance on completely unseen future months — a meaningful result for a temporal extrapolation task.

Actual vs. Predicted

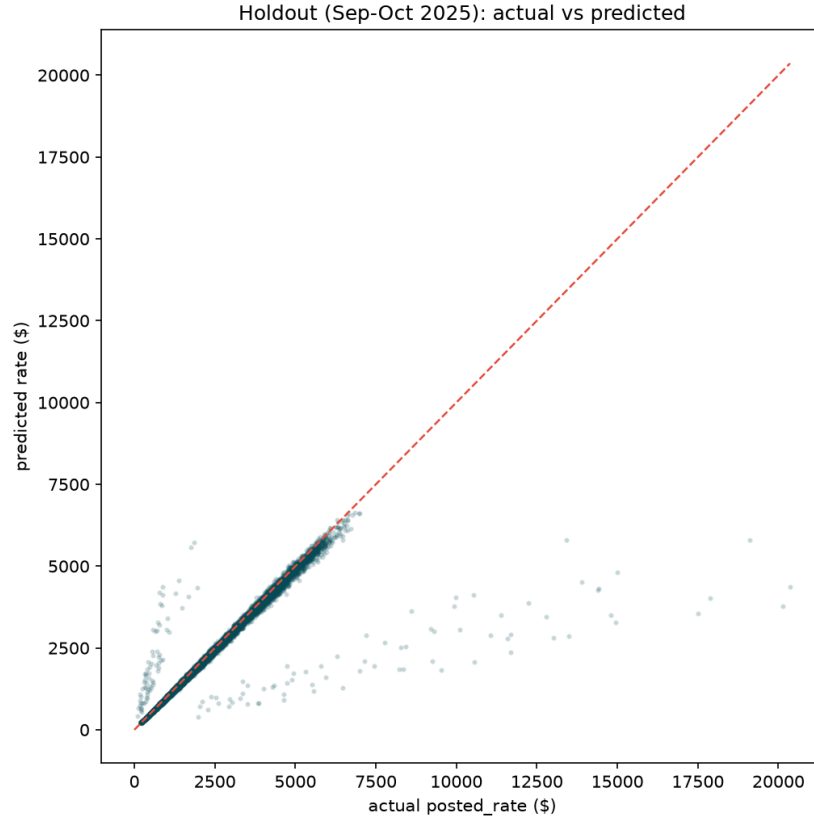


Figure 7: Holdout actual vs. predicted rate (Sep–Oct 2025, $n = 9,523$). The dashed red line is the identity ($\hat{y} = y$). Points cluster tightly along the diagonal for typical loads in the range \$0–\$7,500, confirming strong predictive performance in the primary market segment. High-rate loads (actual $>$ \$10,000) are systematically under-predicted, corresponding to the unexplained spot-market spikes described in Section 2.

Residual Analysis

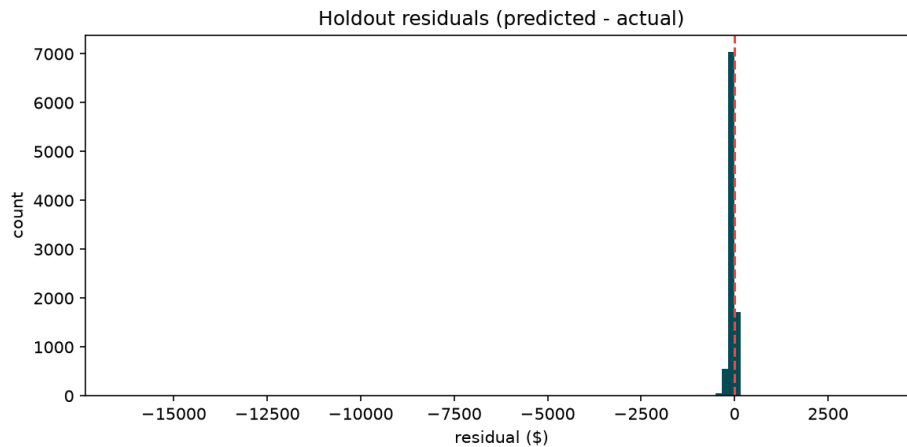


Figure 8: Residual distribution ($\hat{y} - y$) on the holdout set. The distribution is tightly centred on zero (median residual $\approx$ \$0), confirming negligible systematic bias for typical loads. The left tail of large negative residuals corresponds to high-rate outlier loads where the model's conservative L1-trained prediction under-shoots the actual spike value.

Key observations from the residual analysis:

- The residual distribution is highly symmetric and centred near zero for loads in the primary rate range (\$0–\$7,500), confirming the absence of systematic bias across the bulk of the market.
- The left tail (large negative residuals) represents high-rate outlier loads. This is an expected consequence of L1 loss: the model learns to predict near the conditional median, which conservatively under-shoots rare capacity spikes.
- No systematic right-skew in residuals, confirming that the model does not consistently over-predict on any load segment.

Feature Importance

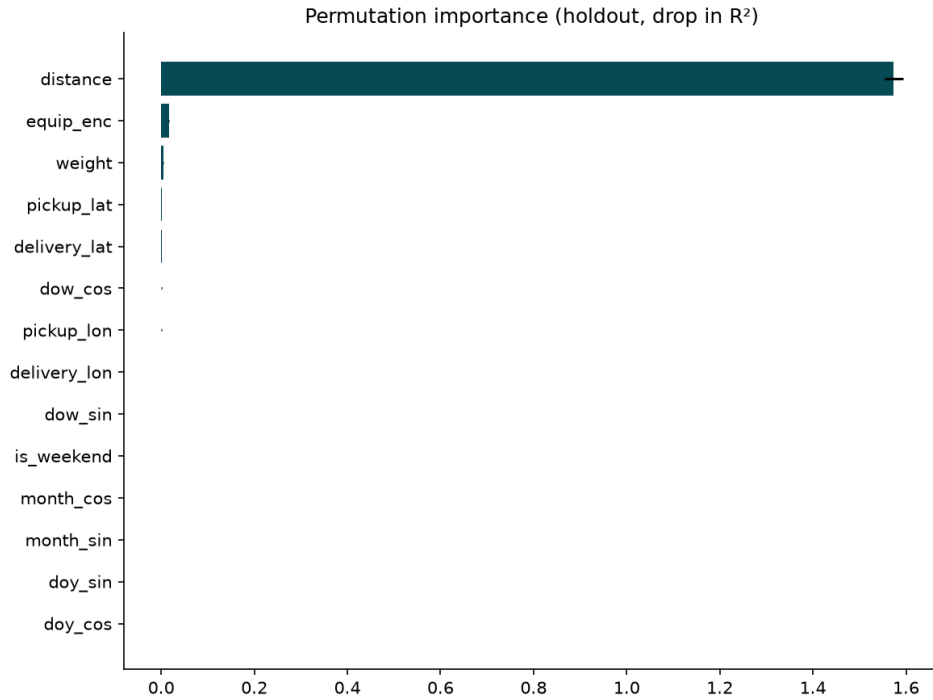


Figure 9: Permutation importance on the Sep–Oct 2025 holdout (mean drop in R^2 when a feature is randomly shuffled across 5 independent repeats; error bars show ± 1 standard deviation). `distance` dominates by a large margin. `equipment` and `weight` contribute meaningfully. Geographic (lat/lon) and date-cyclical features contribute individually very little once the primary three features are known, but collectively enable generalisation to unseen cities and a sensible December date trend.

Observation 1. `distance` accounts for the vast majority of predictive power (permutation importance $\Delta R^2 \approx 1.57$), consistent with its raw Pearson correlation of 0.91.

Observation 2. `equipment` and `weight` show small but non-zero importance, encoding the rate-per-mile premium hierarchy (Reefer > Flatbed > Dry Van) and load-density pricing effects, respectively.

Observation 3. Geographic and date features show near-zero individual permutation importance once `distance`, `equipment`, and `weight` are known. Their value lies in enabling spatial generalisation to new city pairs (lat/lon) and keeping December date values inside the training distribution (cyclical sin/cos encodings), rather than in raw predictive power on seen routes.

December 2025 Rate Forecast

The fixed route is Lexington, KY $\rightarrow$ Fort Wayne, IN, 360 miles, Dry Van, 32,000 lb. Only the date varies across the 31 rows.

Table 7: December 2025 predicted rates for the fixed Lexington → Fort Wayne route (360 mi, Dry Van, 32,000 lb). Minimum and maximum values are in **bold**.

Date (Day)	Rate (\$)	Date (Day)	Rate (\$)
2025-12-01 (Mon)	781.41	2025-12-17 (Wed)	786.20
2025-12-02 (Tue)	790.15	2025-12-18 (Thu)	787.61
2025-12-03 (Wed)	788.84	2025-12-19 (Fri)	780.23
2025-12-04 (Thu)	787.70	2025-12-20 (Sat)	772.75
2025-12-05 (Fri)	783.18	2025-12-21 (Sun)	772.09
2025-12-06 (Sat)	775.15	2025-12-22 (Mon)	772.29
2025-12-07 (Sun)	774.63	2025-12-23 (Tue)	782.58
2025-12-08 (Mon)	775.60	2025-12-24 (Wed)	782.96
2025-12-09 (Tue)	785.15	2025-12-25 (Thu)	785.31
2025-12-10 (Wed)	786.19	2025-12-26 (Fri)	780.96
2025-12-11 (Thu)	787.60	2025-12-27 (Sat)	773.51
2025-12-12 (Fri)	783.27	2025-12-28 (Sun)	773.14
2025-12-13 (Sat)	775.62	2025-12-29 (Mon)	773.28
2025-12-14 (Sun)	774.87	2025-12-30 (Tue)	782.02
2025-12-15 (Mon)	773.96	2025-12-31 (Wed)	783.98
2025-12-16 (Tue)	784.40		
Range: \$772.09 (Dec 21, Sun) to \$790.15 (Dec 2, Tue) — spread $\approx 2.3\%$			
Pattern: weekday-higher / weekend-lower weekly cycle			

Interpretation and Caveats

The predicted range of \$772–\$790 (2.3 % spread across 31 days) reflects the feature importance structure reported in Section 7. For this fixed route where distance, equipment, and weight are all constant, only date features vary, and date features carry near-zero permutation importance in the fitted model (Figure 9).

This forecast should be interpreted as a small residual seasonal effect layered on a rate that is overwhelmingly determined by the fixed route parameters. It is not a high-confidence seasonal trend forecast.

The clear weekday-higher / weekend-lower pattern is a genuine, if small, signal captured from training data (Figure 4). Cyclical sin/cos date encodings (Equation 1) ensure December values fall within the training distribution, avoiding tree-model extrapolation beyond the October training boundary [6].

Reproducibility

All results can be reproduced from the repository root using the following commands. All stochastic components use a fixed random seed (`random_state = 42`).

Table 8: Repository file roles.

File	Role
<code>train.py</code>	Data cleaning, time-based holdout evaluation, final model fit, writes <code>validation_predictions.csv</code> + <code>december_predictions.csv</code>
<code>eda_new.py</code>	Generates all EDA and diagnostic figures in <code>reports/figures/</code> ; trains and compares all five models; produces <code>model_comparison.png</code>
<code>score.py</code>	Validates both output files; generates <code>scorer_results/candidate_december.png</code>
<code>src/features.py</code>	All data cleaning and feature engineering (<code>clean_weight</code> , median imputation, cyclical encodings)
<code>src/model.py</code>	Model definition — HistGBR with absolute-error loss

```
python -m pip install -r requirements.txt
python train.py
python eda_new.py
python score.py --predictions validation_predictions.csv \
                --december-predictions december_predictions.csv
```

Acknowledgements

The author thanks the Spotter team for designing a well-scoped and practically grounded assessment. The dataset captures real-world complexities — temporal extrapolation, unseen geographic entities, heteroscedastic rate distributions, and missing market signals — that make for a genuinely instructive machine learning exercise.

References

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